Micro/Small Business Network Design Framework – Design 2 (OpenWrt One + Netgear)

Reproducible entry-level setup using entirely new hardware. Meets every governance requirement (cable management, guest isolation + QR password + captive basics, VLAN segmentation, protective DNS with NextDNS forced everywhere, IoT isolation, WireGuard VPN with role-based access, power discipline, 3-2-1 config backups) at micro-business scale while staying under $260 total.

Core Philosophy

Same as before: single core device + managed switch for VLANs, patch-panel jumpers only, Velcro cable management, three VLANs (1 Main, 10 Guest, 20 IoT), client isolation on Guest/IoT SSIDs, NextDNS everywhere with firewall DNS blocking, WireGuard VPN limited to approved zones only, no cables on floor or exposed.

Inventory List (Current Approx. Prices March 2026 – Amazon/AliExpress)

Item	Model	Approx. Price	Key Features & Governance Fit	Analysis & Exceptions
Core Router/Firewall	OpenWrt One (Banana Pi official)	$89–$95	Official OpenWRT pre-installed, Wi-Fi 6 dual-band, 2.5GbE + 1GbE ports, full VLAN tagging, WireGuard, client isolation, custom DNS forcing, QoS, firewall rules, WPA3.	Full governance match: native VLAN segmentation, VPN, DNS filtering, guest/IoT isolation. Handles 20–25 devices. No exceptions on features
Managed Switch	Netgear GS108Ev4 (Easy Smart 8-port)	$49–$55	802.1Q VLAN tagging, port trunking, QoS, web UI, fanless	Enables full wired VLAN segmentation + trunk from core. Exact match to policy. No exceptions
Patch Panel	12-port	$22–	Standard	Meets "patch panel +

	Cat6 keystone + short jumpers	$25	wall/rack mount	short jumpers only" rule.
Wall Plates & Cables	5× wall plates + Cat6 + Velcro ties	$40–$45	Removable Velcro only.	Exact cable-management compliance
Power Protection	Listed surge protector or 600VA UPS	$25–$35	Surge + optional battery on dedicated circuits	Meets grounding/surge/UPS rules.

Total Setup Cost: ~$230–$255.

Scales by adding one cheap Wi-Fi 6 AP later if >1,500 sq ft.

Logical & Physical Topology

ISP → OpenWrt One (2.5GbE WAN) → short jumper to patch panel → Netgear switch (trunk port carrying VLANs 1/10/20) → wall plates or additional APs. All excess looped with Velcro; nothing touches floor.

VLAN & Firewall Rules

- VLAN 1: Main LAN
- VLAN 10: Guest (client isolation + QoS limit + DNS filtering)
- VLAN 20: IoT (client isolation + no access to Main)
  Full 802.1Q trunking + firewall blocks all cross-VLAN except explicit VPN allows.

Wireless Guidelines

Built-in Wi-Fi 6 on core + optional second AP. Three SSIDs with client isolation on Guest/IoT. Place for −80 dBm overlap. Guest: WPA3 + physical QR code password.

DNS & Filtering / VPN / Power / Backup

NextDNS forced on all interfaces + firewall drops other DNS. WireGuard server (modern crypto, role-based zones only). All power on surge/UPS. Config exports + 3-2-1 backups (local + cloud).

- 3-2-1 backups (local + cloud).

Exceptions

Only sourcing (AliExpress for $89 lowest price; Amazon version ~$130 still under budget). No functional exceptions — full OpenWRT delivers exact governance match.

Micro/Small Business Network Design Framework – Design 3 (Cudy + Ugreen)

Completely different hardware set. Even lower cost baseline while still delivering strong governance compliance. Ideal for strictest budget or fastest Amazon-only ordering.

Inventory List (Current Approx. Prices March 2026 – Amazon)

Item	Model	Approx. Price	Key Features & Governance Fit	Analysis & Exceptions
Core Router/Firewall	Cudy WR3000 (AX3000 Wi-Fi 6)	$60	5 Gigabit ports, WireGuard server, guest network with isolation, custom DNS, QoS, IPTV/VLAN support, Cudy Mesh ready, APP control	Strong match on guest isolation, VPN, DNS forcing, QoS. Wi-Fi 6 covers 20–25 devices.
Managed Switch	Ugreen 8-port Gigabit with VLAN mode	$35–$40	One-touch VLAN isolation mode + standard/extend, metal fanless, QoS basics.	Provides port-level isolation for wired Guest/IoT without cross-talk. Enables segmentation via switch + firewall rules.
Patch Panel	12-port Cat6 keystone + short jumpers	$22–$25	Standard	Meets patch-panel rule.
Wall Plates & Cables	5× wall plates + Cat6 + Velcro ties	$40–$45	Velcro only.	Exact cable compliance

	Velcro ties			
Power Protection	Listed surge protector or 600VA UPS	$25–$35	Surge + battery option	Meets power policy

Total Setup Cost: ~$190–$215 (lowest of the three designs).

Logical & Physical Topology

ISP → Cudy WAN → patch panel jumpers → Ugreen switch (VLAN-isolation ports) → wall plates. Same professional cabling rules.

VLAN & Firewall Rules

- VLAN 1 / Main zone on dedicated ports
- Guest & IoT isolated via switch VLAN mode + Cudy firewall rules (no cross-talk).
  Matches segmentation policy via isolation + access controls.

Wireless Guidelines

Built-in AX3000 Wi-Fi 6 + three SSIDs with client isolation on Guest/IoT. QR-code password sharing. Same Ekahau placement rules.

DNS & Filtering / VPN / Power / Backup

NextDNS forced + DNS blocking at firewall. WireGuard with role-based access. Surge/UPS power. 3-2-1 config backups.

Exceptions (Documented for Leadership)

1. Full 802.1Q trunk tagging not native in stock firmware (use switch VLAN-isolation mode + firewall rules instead — identical approach to your provided DMZ-exception document).
2. Optional one-time OpenWRT flash for deeper VLAN tagging if future growth requires it (simple web-based process, fully documented in deployment guide).
   Both keep the setup entry-level replicable and under $215 while still isolating Guest/IoT exactly as required.

Both new designs are 100% hardware-distinct from the previous GL.iNet/TP-Link stack. They remain fully reproducible by any entry-level tech using the exact shopping lists above. Design 2 (OpenWrt One) gives zero-compromise OpenWRT power; Design 3 (Cudy) gives the absolute cheapest Amazon-only path with documented, governance-aligned exceptions.

All three designs (including the original) share the same generic cabling/power rules and NextDNS/backup/VPN policies so leadership can standardize on one framework.

Approve any of the three (or mix core + switch) and I will deliver the matching step-by-step deployment guide + exact OpenWRT/Cudy config templates + wiring diagram for audit. Ready when you are.